



Topic 2: Data and Signals

CSE 4405: Data and Telecommunications

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Chapter 2: Data and Signals

Opening statement

Information can be voice, image, numeric data, characters, code, picture, and so on. To be transmitted, information must be converted into electromagnetic signals.

What this chapter studies

- Data and signal types
- Time and frequency domains
- Bandwidth
- Digital signaling
- Transmission impairment
- Performance limits

2.1 Analog and Digital

Data can be analog or digital

The term **analog data** refers to information that is continuous; **digital data** refers to information that has discrete states. Analog data take on continuous values. Digital data take on discrete values.

Important separation

Data answers what the information is.

Signal answers how that information is represented during transmission.

Topics discussed in this section

- Analog and digital data
- Analog and digital signals
- Periodic and nonperiodic signals

Analog and Digital Signals

Analog signal

Having **infinitely many levels of intensity** over a period of time. As the wave moves from one value to another, it passes through and includes an infinite number of values along its path.

Digital signal

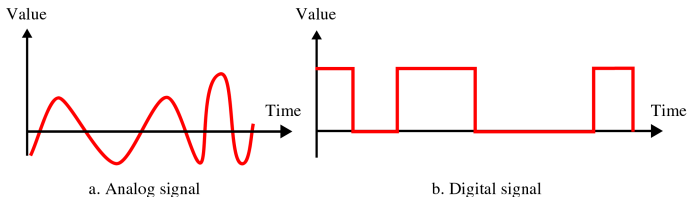
Having only a **limited number of defined values**. A simple binary signal may use one level for 1 and another level for 0.

Key idea

A signal may be continuous in amplitude or restricted to levels. That distinction is separate from whether the underlying data itself is analog or digital.

Comparison of Analog and Digital Signals

- **Analog:** infinitely many values in a range
- **Digital:** a limited set of levels
- The horizontal axis is **time**; the vertical axis is **value**
- Both are legitimate signal representations



Periodic and Aperiodic Signals

Periodic signals

A periodic signal consists of a **continuously repeated pattern**. The **period** of a signal, T , is expressed in seconds, and a **cycle** is one full pattern.

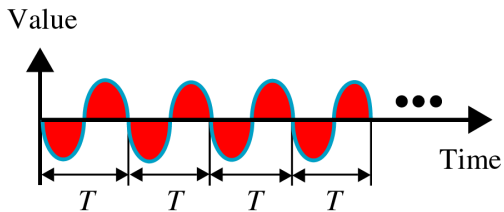
Aperiodic signals

An aperiodic signal changes constantly without exhibiting a pattern or cycle that repeats over time. In other words, the signal has **no repetitive pattern**.

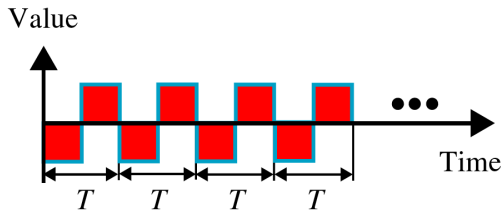
Periodic signals repeat. Aperiodic signals vary without repetition.

Example of Periodic Signals

Figure: Example periodic waveforms



a. Analog



b. Digital

The figure contrasts analog periodic oscillation with a periodic digital pattern whose cycle length is T .

Signals Used in Data Communication

Common engineering choice

In data communication, we commonly use **periodic analog signals** and **aperiodic digital signals**.

- Periodic analog signals are natural for carrier-based analog transmission.
- Aperiodic digital signals are natural for data transmission because bit patterns rarely repeat forever.
- The same chapter therefore studies both **repeating waveforms** and **one-shot bit patterns**.

Simple and Composite Periodic Analog Signals

Classification

Periodic analog signals can be classified as **simple** or **composite**.

- A **simple** periodic analog signal is a **sine wave**; it cannot be decomposed into simpler signals.
- A **composite** periodic analog signal is composed of multiple sine waves.

Foundation

The sine wave is the most fundamental form of a periodic analog signal.

Topics discussed in Section 2.2

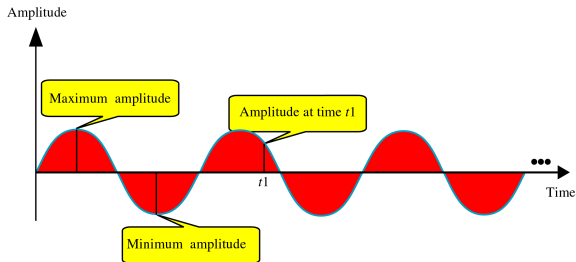
- Sine wave
- Wavelength
- Time and frequency domain
- Composite signals
- Bandwidth

Sine Wave

Definition

A sine wave is the simplest periodic signal. A sine wave can be fully described by three characteristics.

1. **Amplitude**
2. **Period** or **frequency**
3. **Phase**



Amplitude, Period, and Frequency

Amplitude

Amplitude refers to the **height** of the signal. At a specific instant, it is the signal value itself. Common units are **volts**, **amperes**, and **watts**.

Period and frequency

Period is the amount of time, in seconds, that a signal needs to complete one cycle. **Frequency** is the number of periods made in one second.

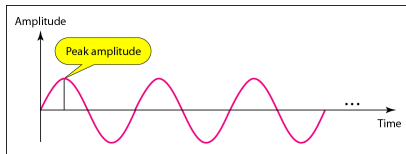
Fundamental relationship

$$f = \frac{1}{T}, \quad T = \frac{1}{f}$$

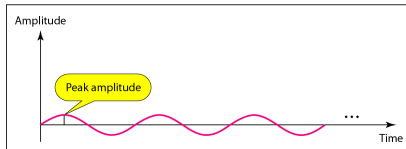
Unit of frequency: hertz (Hz). Unit of period: seconds (s).

Figure 2.3: Same Phase and Frequency, Different Amplitudes

Figure: Two signals with the same phase and frequency but different amplitudes



a. A signal with high peak amplitude

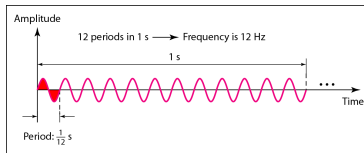


b. A signal with low peak amplitude

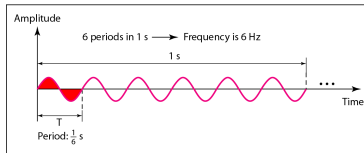
Only the height changes; the cycle timing and phase relationship stay fixed.

Figure 2.4: Same Amplitude and Phase, Different Frequencies

Figure: Two signals with the same amplitude and phase but different frequencies



a. A signal with a frequency of 12 Hz



b. A signal with a frequency of 6 Hz

Higher frequency means more cycles per second and therefore a shorter period.

Table 2.1: Units of Period and Frequency

Quantity	Common units	Reading the unit
Frequency	1 Hz, 1 kHz = 10^3 Hz, 1 MHz = 10^6 Hz, 1 GHz = 10^9 Hz	Cycles per second
Period	1 s, 1 ms = 10^{-3} s, 1 μ s = 10^{-6} s, 1 ns = 10^{-9} s	Time required for one cycle

Reading the table

A signal with $T = 0.5$ s has $f = 2$ Hz. A signal with $f = 1$ MHz has $T = 1$ μ s.

More About Frequency

Frequency is the rate of change with respect to time

A change in a short span of time means **high frequency**. A change in a long span of time means **low frequency**.

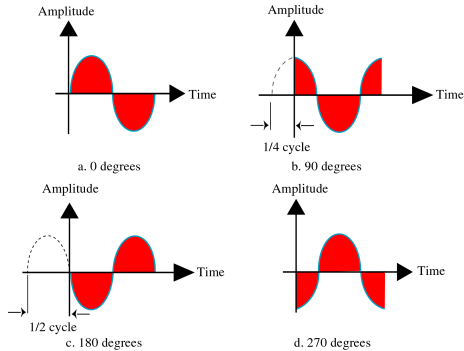
- If a signal does not change at all, its frequency is **zero**.
- If a signal changes instantaneously, its frequency is **infinity**.
- Frequency therefore tells us how rapidly the waveform evolves.

Phase

Phase describes the position of the waveform relative to time zero.

- At $t = 0$, phase tells us where the signal sits within its cycle.
- A positive or negative phase term shifts the waveform relative to a reference sine wave.
- Phase is usually measured in **degrees** or **radians**.

Relationship Between Different Phases



Signals with the same amplitude and frequency may still differ because one waveform is shifted relative to another.

Example 2.6: Phase Offset

Problem

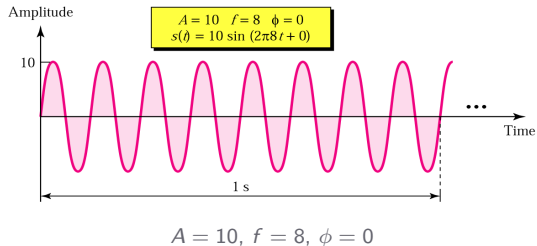
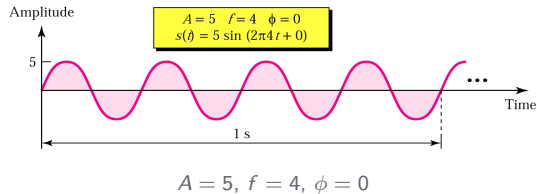
A sine wave is offset one-sixth of a cycle with respect to time zero. What is its phase in degrees and radians?

$$\begin{aligned}\frac{1}{6} \text{ cycle} &= \frac{1}{6} \times 360^\circ = 60^\circ \\ 60^\circ &= 60 \times \frac{2\pi}{360} = \frac{\pi}{3} \approx 1.046 \text{ rad}\end{aligned}$$

Interpretation

A full cycle is 360° or 2π radians. One-sixth of that cycle is therefore 60° or $\pi/3$ radians.

Sine-Wave Examples

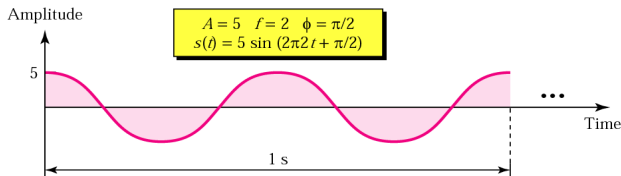


The figure pair shows how increasing **amplitude** changes the height of the signal and increasing **frequency** changes the number of oscillations in the same time interval.

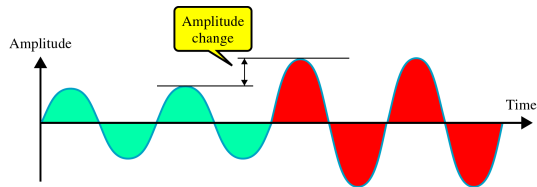
Another Sine-Wave Example

Phase-shifted example

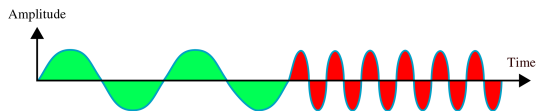
Here $A = 5$, $f = 2$, and $\phi = \pi/2$. The waveform starts from a different position because the phase term shifts the signal relative to time zero.



Amplitude Change and Frequency Change



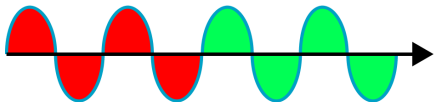
Amplitude change



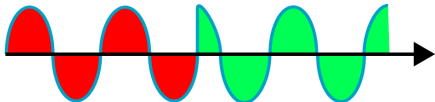
Frequency change

Amplitude change alters signal strength. **Frequency change** alters how quickly the waveform oscillates.

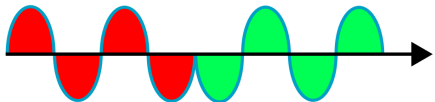
Phase Change



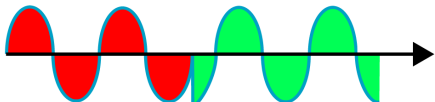
a. No phase change



b. 90 degree phase change



c. 180 degree phase change



d. 270 degree phase change

No phase change, 90° , 180° , and 270° phase changes produce horizontal shifts of the waveform.

Wavelength

Definition

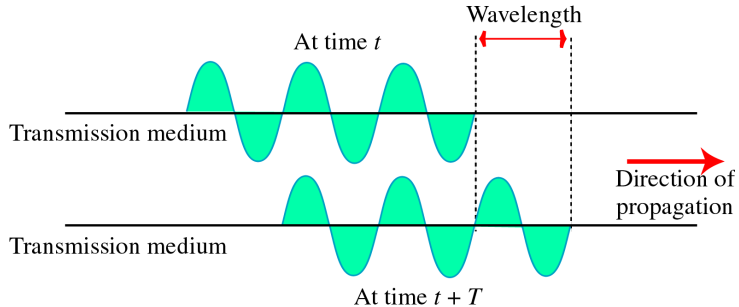
Wavelength is the distance a signal travels in one period.

$$\lambda = \frac{c}{f}$$

Meaning: wavelength = propagation speed / frequency

- $c \approx 3 \times 10^8$ m/s in vacuum
- Lower frequency means longer wavelength
- Higher frequency means shorter wavelength

Wavelength Illustration



Lower-frequency waves occupy longer distances; higher-frequency waves occupy shorter distances.

Time versus Frequency Domain

Time domain

The time domain shows **instantaneous amplitude** with respect to **time**.

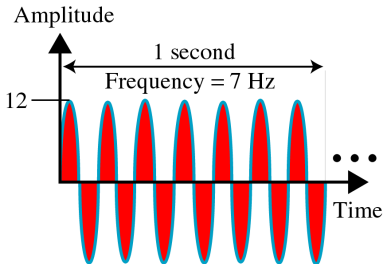
Frequency domain

The frequency domain shows **maximum amplitude** with respect to **frequency**.

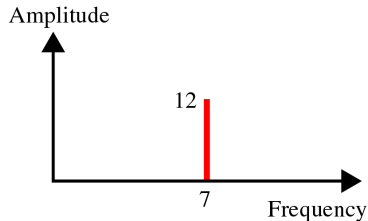
Two views of one signal

The same signal can be described as a waveform changing over time or as a set of frequency components.

Time and Frequency Domains



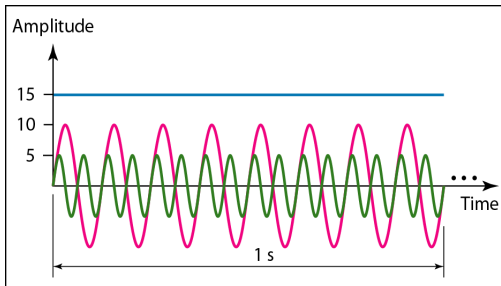
a. Time domain



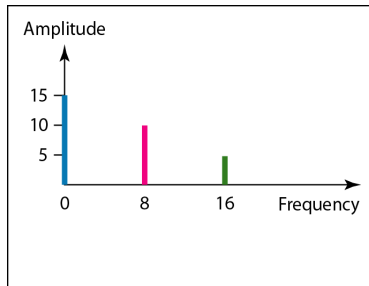
b. Frequency domain

The peak value of a sine wave in time maps to a single spectral line at its frequency in the frequency domain.

Figure 2.8: Time Domain and Frequency Domain of Three Sine Waves



a. Time-domain representation of three sine waves with frequencies 0, 8, and 16



b. Frequency-domain representation of the same three signals

A complete sine wave in the time domain is represented by one single spike in the frequency domain.

Composite Signals and Fourier Analysis

Why a single sine wave is not enough

A single-frequency sine wave is not useful in data communications; we need to change one or more of its characteristics to make it useful. When we do that, the result becomes a **composite signal**.

Fourier analysis

Any composite signal is a combination of simple sine waves with different **frequencies, amplitudes, and phases**. If the composite signal is periodic, the decomposition yields **discrete** frequencies; if it is nonperiodic, the decomposition yields **continuous** frequencies.

Example 2.8: A Periodic Composite Signal

Source description

Figure 2.9 shows a periodic composite signal with frequency f . The book compares it to three alarm systems, each with a different frequency, so that we can see how a periodic composite signal decomposes.

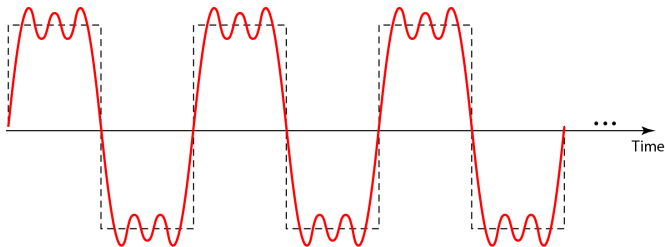
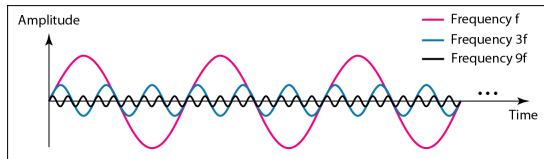
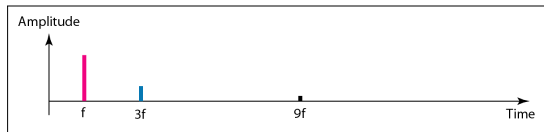


Figure 2.10: Decomposition in Time and Frequency



a. Time-domain decomposition of a composite signal



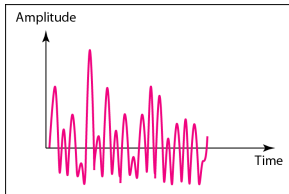
b. Frequency-domain decomposition of the composite signal

The figure separates a periodic composite signal into simple components in the time domain and the matching discrete spectral lines in the frequency domain.

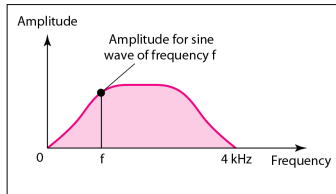
Example 2.9: A Nonperiodic Composite Signal

Source description

Figure 2.11 shows a nonperiodic composite signal such as the signal created by a microphone or telephone when a word or two is pronounced. It is nonperiodic because speech is not repeated with exactly the same tone forever.



a. Time domain



b. Frequency domain

A Demonstration on Fourier

Recommended demonstration

The original slides point to an online Fourier demonstration that lets you build a complex waveform by adding harmonics.

<http://www.earlevel.com/Digital%20Audio/harmonigraf.html>

Why it helps

Fourier analysis is easier to remember when you can visually add harmonics and watch the composite waveform change.

Frequency Spectrum and Bandwidth

Frequency spectrum

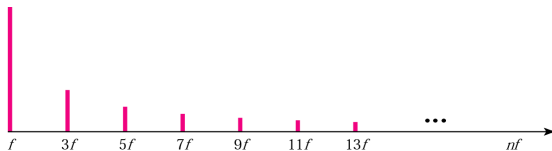
The frequency spectrum of a signal is the collection of sine-wave components that make the original signal.

Bandwidth

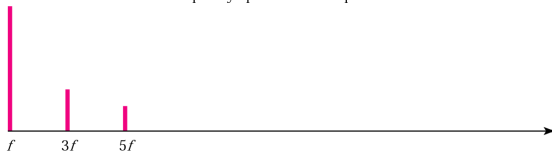
For a composite signal, bandwidth is the difference between its highest and lowest frequencies.

$$\text{BW} = f_{\max} - f_{\min}$$

Frequency Spectrum



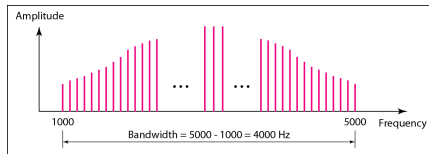
a. Frequency spectrum of a square wave



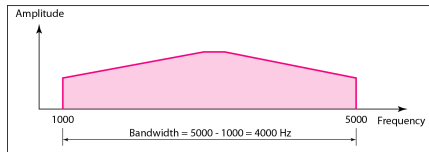
b. Frequency spectrum of an approximation with only three harmonics

The frequency spectrum shows the amplitude of each component frequency.

Figure 2.12: Bandwidth of Periodic and Nonperiodic Composite Signals



a. Bandwidth of a periodic signal



b. Bandwidth of a nonperiodic signal

Periodic composite signals occupy discrete frequencies; nonperiodic composite signals occupy a continuous band.

Example 2.10: Bandwidth of a Periodic Signal

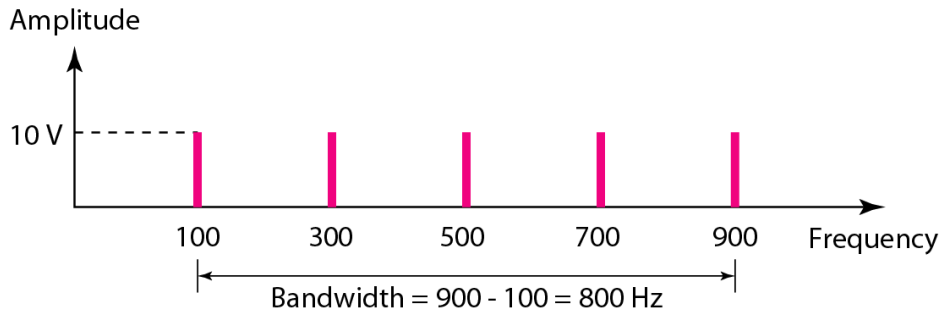
Problem

A periodic signal is decomposed into five sine waves with frequencies 100, 300, 500, 700, and 900 Hz. If all components have peak amplitude 10 V, draw the spectrum and find the bandwidth.

$$f_{\max} = 900 \text{ Hz}, \quad f_{\min} = 100 \text{ Hz}$$

$$\text{BW} = 900 - 100 = 800 \text{ Hz}$$

Figure 2.13: Bandwidth for Example 2.10



Example 2.11: Lowest Frequency from Bandwidth

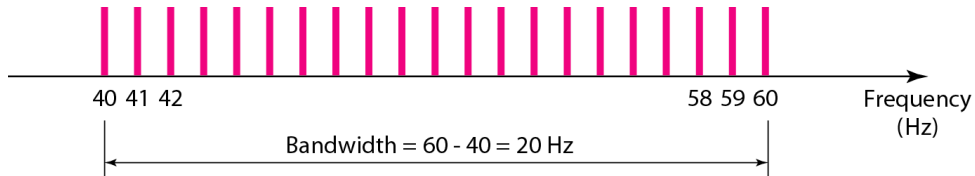
Problem

A periodic signal has bandwidth 20 Hz. The highest frequency is 60 Hz. Find the lowest frequency and draw the spectrum if all components have the same amplitude.

$$BW = f_{\max} - f_{\min}$$

$$20 = 60 - f_{\min} \Rightarrow f_{\min} = 40 \text{ Hz}$$

Figure 2.14: Bandwidth for Example 2.11



Example 2.12: Bandwidth of a Nonperiodic Signal

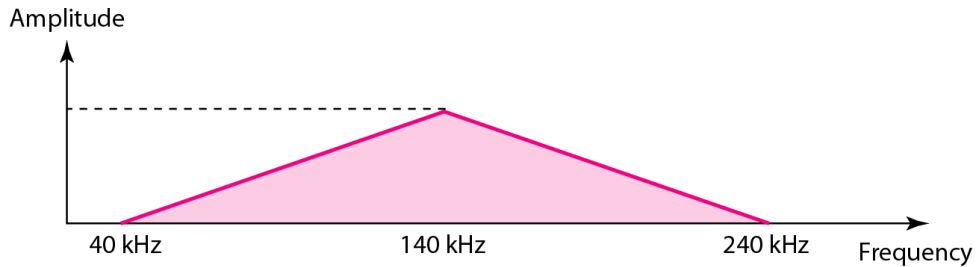
Problem

A nonperiodic composite signal has bandwidth 200 kHz, middle frequency 140 kHz, and peak amplitude 20 V. The two extreme frequencies have amplitude 0. Draw its frequency-domain representation.

$$f_{\min} = 140 - 100 = 40 \text{ kHz}$$

$$f_{\max} = 140 + 100 = 240 \text{ kHz}$$

Figure 2.15: Bandwidth for Example 2.12



2.3 Digital Signals

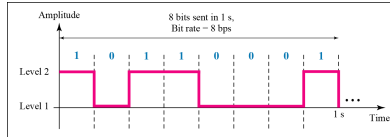
Section opening

In addition to being represented by an analog signal, information can also be represented by a **digital signal**. A 1 may be encoded as a positive voltage and a 0 as zero voltage. A digital signal can have more than two levels; in that case, we can send more than 1 bit per level.

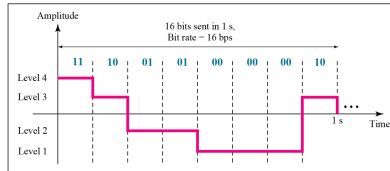
Topics discussed in this section

- Bit rate
- Bit length
- Digital signal as a composite analog signal
- Transmission of digital signals

Figure 2.16: Two Digital Signals



a. A digital signal with two levels



b. A digital signal with four levels

With more signal levels, more bits can be conveyed during the same time interval.

Example 2.16: Bits Needed per Signal Level

Problem

A digital signal has eight levels. How many bits are needed per level?

$$\text{bits per level} = \log_2 L = \log_2 8 = 3$$

Interpretation

Each signal level is represented by **3 bits**. More levels increase the number of bits conveyed per signaling symbol.

Example 2.19: Bit Rate of a Digitized Voice Channel

Problem

A digitized voice channel is created from a 4-kHz analog voice signal. We sample at twice the highest frequency and use 8 bits per sample. What is the required bit rate?

$$\text{sampling rate} = 2 \times 4,000 = 8,000 \text{ samples/s}$$

$$\text{bit rate} = 8,000 \times 8 = 64,000 \text{ bps} = 64 \text{ kbps}$$

Example 2.20: HDTV Bit Rate

Problem

What is the bit rate for high-definition TV (HDTV)?

Given

- Screen ratio: 16 : 9
- Resolution: 1920×1080 pixels
- Refresh rate: 30 screens per second
- Color depth: 24 bits per pixel

$$1920 \times 1080 \times 30 \times 24 \approx 1.5 \text{ Gbps}$$

Exact raw rate: 1,492,992,000 bps

Practical note

Television systems reduce this raw rate to roughly 20 to 40 Mbps through compression.

Bit Length

Bit length

Bit length is the distance one bit occupies on the transmission medium.

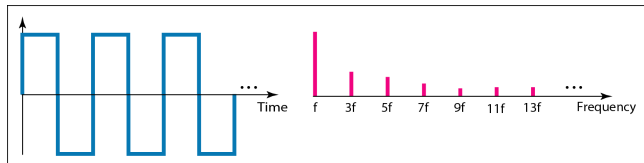
$$\text{Bit length} = v \times T_b$$

Meaning: propagation speed $\times$ bit duration

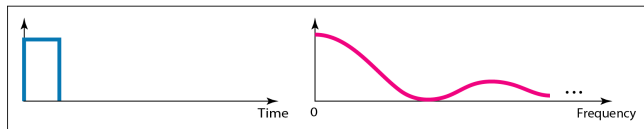
Physical meaning

A higher propagation speed or longer bit duration stretches the bit over a longer physical distance.

Digital Signal as a Composite Analog Signal



a. Time and frequency domains of periodic digital signal



b. Time and frequency domains of nonperiodic digital signal

A periodic digital signal has discrete harmonics; a nonperiodic digital signal has a continuous frequency-domain representation.

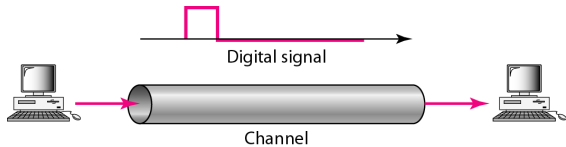
Transmission of Digital Signals

Two transmission types

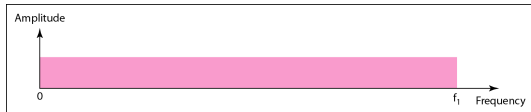
Digital signals are transmitted as **baseband** or **broadband**. A digital signal is a composite analog signal with an **infinite bandwidth**.

Practical consequence

Real channels never offer infinite bandwidth, so every physical medium passes only an approximation of the ideal digital waveform.



Low-Pass Channels and Baseband Transmission



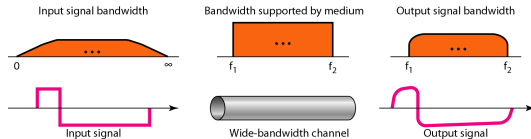
a. Low-pass channel, wide bandwidth



b. Low-pass channel, narrow bandwidth

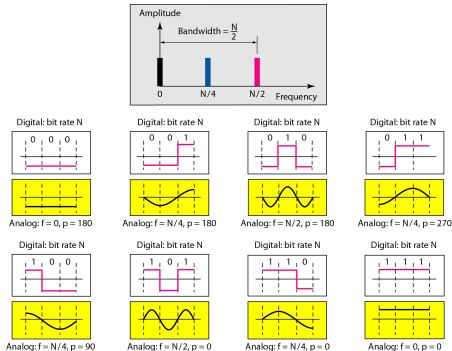
Bandwidths of two low-pass channels

Baseband transmission that preserves the digital shape is possible only if the low-pass channel has **infinite** or **very wide bandwidth**.



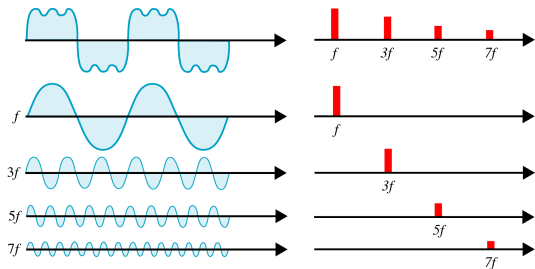
Baseband transmission using a dedicated medium

Low-Pass Channel with Limited Bandwidth



The first harmonic gives a rough approximation of the digital signal in the worst case.

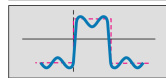
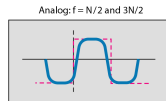
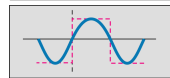
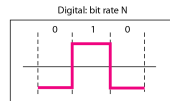
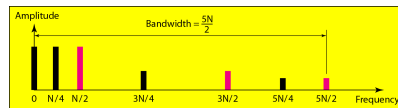
Simulating a Digital Signal with Harmonics



a. Time domain

b. Frequency domain

Complex waveform



Analog: $f = N/2$

Analog: $f = N/2, 3N/2, \text{ and } 5N/2$

Simulation

using the first three harmonics

In baseband transmission, the required bandwidth is proportional to the bit rate: if we need to send bits faster, we need more bandwidth.

Table 2.2: Bandwidth Requirements

Baseband rule of thumb

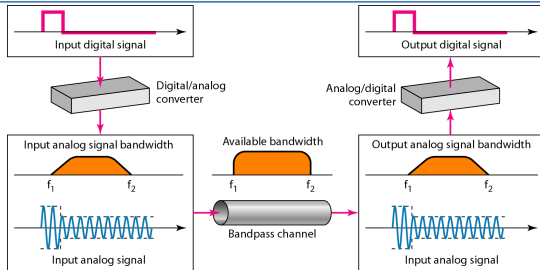
For a digital signal with bit rate n , preserving more harmonics requires more bandwidth.

Bit rate	Harmonic 1	Harmonics 1, 3	Harmonics 1, 3, 5
$n = 1$ kbps	$B = 500$ Hz	$B = 1.5$ kHz	$B = 2.5$ kHz
$n = 10$ kbps	$B = 5$ kHz	$B = 15$ kHz	$B = 25$ kHz
$n = 100$ kbps	$B = 50$ kHz	$B = 150$ kHz	$B = 250$ kHz

Broadband Transmission Using Modulation

Bandpass channel

If the available channel is a **bandpass channel**, we cannot send the digital signal directly to the channel. We need to convert the digital signal to an analog signal before transmission.



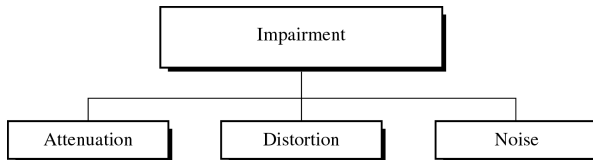
The digital signal is translated to an analog waveform that fits within the available passband.

2.4 Transmission Impairment

Impairment idea

Transmission media are not perfect. The signal at the beginning and end of the medium is not the same because the medium introduces impairment.

- **Attenuation**
- **Distortion**
- **Noise**



Attenuation and Decibels

Attenuation

Attenuation means **loss of energy**. When a signal travels through a medium, it loses some of its energy. To compensate for this loss, amplifiers are used.

$$\text{dB} = 10 \log_{10} \left(\frac{p_2}{p_1} \right)$$

Reading the sign

A negative dB value means power has been reduced. A positive dB value means power has been increased.

Decibel Examples

Power reduced to one-half

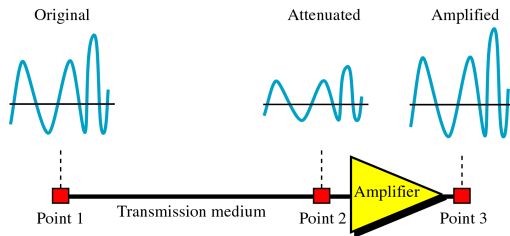
$$p_2 = \frac{1}{2} p_1$$

$$10 \log_{10} \left(\frac{0.5 p_1}{p_1} \right) = 10 \log_{10}(0.5) \approx -3 \text{ dB}$$

Power increased ten times

$$p_2 = 10 p_1$$

$$10 \log_{10} \left(\frac{10 p_1}{p_1} \right) = 10 \log_{10}(10) = 10 \text{ dB}$$



Point-4 example

If the cumulative changes are -3 dB , then $+7 \text{ dB}$, then -3 dB , the total at point 4 is

$$-3 + 7 - 3 = +1 \text{ dB.}$$

Example 2.29: Power from dBm

Problem

Sometimes the decibel is used to measure signal power in milliwatts. In this case it is called **dBm** and is calculated as

$$\text{dBm} = 10 \log_{10} P_m,$$

where P_m is the power in milliwatts. Calculate the power of a signal with $\text{dBm} = -30$.

$$-30 = 10 \log_{10} P_m \Rightarrow \log_{10} P_m = -3 \Rightarrow P_m = 10^{-3} \text{ mW}$$

Equivalent power

$$10^{-3} \text{ mW} = 10^{-6} \text{ W} = 1 \mu\text{W}.$$

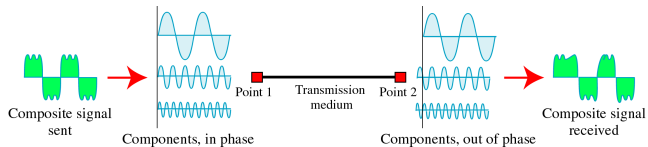
Distortion

Definition

Distortion means that the signal changes its **form** or **shape**.

Reason

Different frequency components may travel with different delays, so the components no longer line up in phase at the receiver.

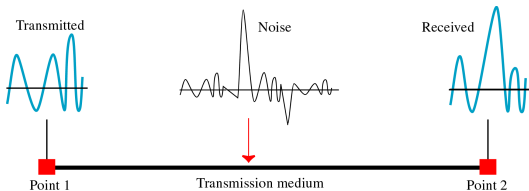


Noise Types

Noise

Noise is **unwanted energy** added to the signal.

- **Thermal noise:** random motion of electrons
- **Induced noise:** motors, appliances, or other electrical sources
- **Crosstalk:** the effect of one wire or channel on another
- **Impulse noise:** spikes from power lines, lightning, and similar events



Signal-to-Noise Ratio

Definitions

The signal-to-noise ratio compares average signal power with average noise power.

$$\text{SNR} = \frac{\text{average signal power}}{\text{average noise power}}$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(\text{SNR})$$

Example 2.31: Computing SNR

Problem

If the signal power is 10 mW and the noise power is 1 μ W, find SNR and SNR_{dB}.

$$\text{SNR} = \frac{10 \text{ mW}}{1 \mu\text{W}} = \frac{10,000 \mu\text{W}}{1 \mu\text{W}} = 10,000$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10}(10,000) = 40 \text{ dB}$$

Example 2.32 and Figure 2.30

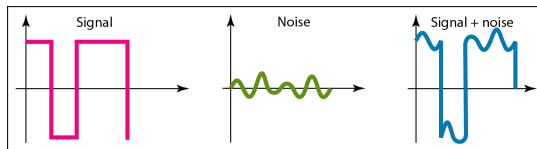
Example 2.32

For a noiseless channel,

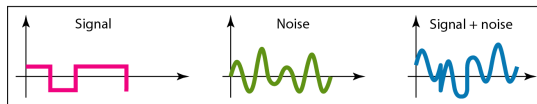
$$\text{SNR} = \frac{\text{signal power}}{0} = \infty$$

$$\text{SNR}_{\text{dB}} = 10 \log_{10} \infty = \infty.$$

We can never achieve this ratio in real life; it is an ideal.



a. Large SNR



b. Small SNR

Data Rate Depends on Three Factors

Three factors

Data rate depends on

1. the available bandwidth,
2. the number of signal levels we can use, and
3. the quality of the channel (its noise level).

Reliability trade-off

Increasing the number of signal levels may reduce the reliability of the system.

Nyquist Bit Rate

Noiseless channel

For a noiseless channel, the Nyquist bit-rate formula gives the theoretical maximum rate.

$$\text{Bit rate} = 2 \times \text{Bandwidth} \times \log_2 L$$

Meaning of L

L is the number of signal levels.

Example 2.34: Nyquist Bit Rate

Problem

For a noiseless channel with bandwidth 3000 Hz and $L = 2$, find the maximum bit rate.

$$\text{Bit rate} = 2 \times 3000 \times \log_2 2 = 6000 \text{ bps}$$

Shannon Capacity

Noisy channel

For a noisy channel, the Shannon formula gives the theoretical upper limit:

$$C = B \log_2(1 + \text{SNR})$$

Example 2.37

Consider an extremely noisy channel in which the signal-to-noise ratio is almost zero. Then

$$C = B \log_2(1 + 0) = B \log_2 1 = 0.$$

If the signal is completely buried in noise, no information can be carried reliably.

Examples 2.39 and 2.40

Example 2.39

If $\text{SNR}_{\text{dB}} = 36$ and the bandwidth is 2 MHz,

$$\text{SNR} = 10^{36/10} = 10^{3.6} \approx 3981$$

$$C = 2 \times 10^6 \log_2(1 + 3981) \approx 23.9 \text{ Mbps.}$$

Example 2.40

For practical purposes, when SNR is very high,

$$1 + \text{SNR} \approx \text{SNR}$$

so

$$C \approx B \log_2(\text{SNR}).$$

For the previous example, the high-SNR approximation gives almost the same capacity.

Example 2.41: Shannon Upper Limit

Problem

We have a channel with 1-MHz bandwidth and $\text{SNR} = 63$. What are appropriate bit rate and signal levels?

$$C = 10^6 \log_2(1 + 63) = 10^6 \log_2 64 = 6 \text{ Mbps}$$

Step 1: The Shannon formula gives the upper limit.

Example 2.41: Choosing Signal Levels

Step 2: choose a practical rate

For better performance, choose a lower rate such as 4 Mbps instead of the 6 Mbps upper limit.

$$4 \text{ Mbps} = 2 \times 1 \text{ MHz} \times \log_2 L$$

$$\log_2 L = 2 \Rightarrow L = 4$$

Takeaway

Shannon tells us the upper limit; Nyquist tells us how many signal levels are needed for a chosen rate below that limit.

2.6 Performance

Section opening

One important issue in networking is performance: **how good is the network?** This section introduces the terms needed in later chapters.

Topics discussed in this section

- Bandwidth
- Throughput
- Latency (delay)
- Bandwidth-delay product

Definition of Bandwidth

Two meanings of bandwidth

In networking, the word **bandwidth** is used in two contexts.

- **Bandwidth in hertz:** the range of frequencies in a composite signal or the range of frequencies that a channel can pass.
- **Bandwidth in bits per second:** the speed of bit transmission in a channel or link.

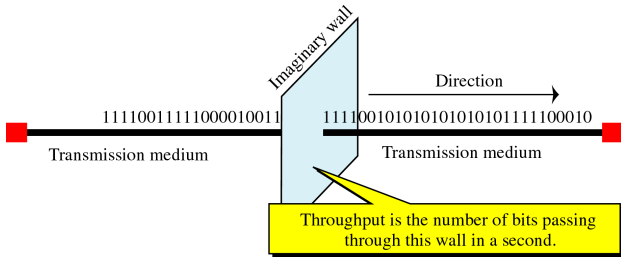
Keep the unit in mind

If the unit is **Hz**, bandwidth is a frequency range. If the unit is **bps**, bandwidth is a data-transmission rate.

Throughput

Definition

Throughput is the measurement of how fast data can pass through a point.



Example 2.44: Throughput Calculation

Problem

A network with bandwidth 10 Mbps carries an average of 12,000 frames per minute, each with 10,000 bits. What is the throughput?

$$\frac{12,000}{60} = 200 \text{ frames/s}$$

$$200 \times 10,000 = 2,000,000 \text{ bps} = 2 \text{ Mbps}$$

Observation

The throughput is almost one-fifth of the nominal bandwidth in this case.

Latency (Delay)

Definition

Latency or delay is how long it takes for an entire message to completely arrive at the destination from the time the first bit is sent from the source.

$$\text{Latency} = \text{propagation time} + \text{transmission time} + \text{queuing time} + \text{processing delay}$$

- **Propagation time:** Time for a bit to travel from source to destination
- **Transmission time:** Time needed to put all bits of a packet onto the link.

Propagation Time

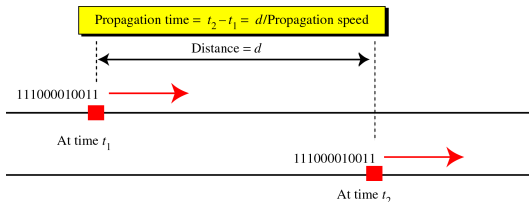
Formula

$$T_p = \frac{d}{v}$$

Example 2.45

For $d = 12,000$ km and propagation speed 2.4×10^8 m/s,

$$T_p = \frac{12,000 \times 10^3}{2.4 \times 10^8} = 50 \text{ ms}$$



This example shows that a bit can cross the Atlantic Ocean in only 50 ms if a direct cable connects the two points.

Transmission Time

Transmission time

$$T_t = \frac{\text{message size}}{\text{bandwidth}}$$

Example 2.46

What are the propagation time and the transmission time for a 2.5-kbyte message if the bandwidth is 1 Gbps, the distance is 12,000 km, and the propagation speed is 2.4×10^8 m/s?

$$\text{message size} = 2,500 \times 8 = 20,000 \text{ bits}$$

$$T_t = \frac{20,000}{10^9} = 20 \mu s$$

Example 2.46: Delay Comparison

$$T_p = \frac{12,000 \times 10^3}{2.4 \times 10^8} = 50 \text{ ms}$$
$$50 \text{ ms} \gg 20 \text{ } \mu\text{s}$$

Dominant delay

Because the message is short and the bandwidth is high, the dominant factor is **propagation time**, not the transmission time. The transmission time can be ignored.

Bandwidth-Delay Product

Definition

Bandwidth-delay product (BDP) means the amount of data that can be “in the link” at one time.

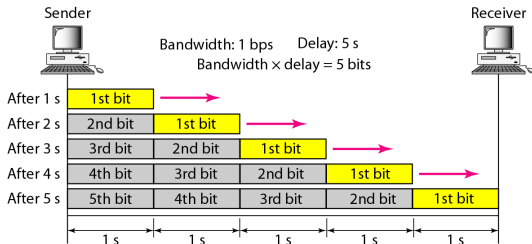


Figure 2.31: filling the link with bits, case 1

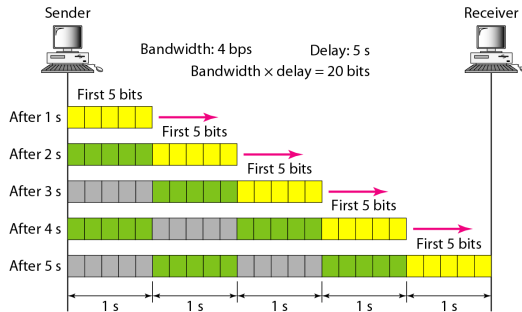


Figure 2.32: filling the link with bits, case 2

Example 2.48: Concept of Bandwidth-Delay Product

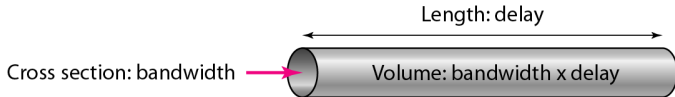
Pipe analogy

We can think of the link between two points as a pipe. The cross section of the pipe represents bandwidth, the length of the pipe represents delay, and the volume represents the bandwidth-delay product.

$$\text{bits in link} = B \times T_p$$

$$10^9 \times 0.05 = 50 \times 10^6 \text{ bits}$$

$$= 6.25 \text{ MB}$$



Summary (1)

Data and signals

- Data must be transformed to electromagnetic signals to be transmitted.
- Data can be analog or digital. Analog data are continuous; digital data have discrete states.
- Signals can be analog or digital. Analog signals can have an infinite number of values; digital signals have a limited number of values.

Periodic signals and phase

- In data communications, we commonly use periodic analog signals and nonperiodic digital signals.
- Phase describes the position of the waveform relative to time 0.
- A complete sine wave in the time domain can be represented by one single spike in the frequency domain.

Summary (2)

Composite signals

- A single-frequency sine wave is not useful in data communications; we need a composite signal.
- According to Fourier analysis, any composite signal is a combination of simple sine waves with different frequencies, amplitudes, and phases.
- The bandwidth of a composite signal is the difference between its highest and lowest frequencies.

Digital transmission

- A digital signal is a composite analog signal with an infinite bandwidth.
- Baseband transmission that preserves digital shape requires a low-pass channel with infinite or very wide bandwidth.
- If the available channel is bandpass, the digital signal must be converted to an analog signal before transmission.

Summary (3)

Transmission and limits

- If the available channel is bandpass, the digital signal must be converted to an analog signal before transmission.
- For a noiseless channel, the Nyquist formula defines the theoretical maximum bit rate.
- For a noisy channel, the Shannon formula gives the maximum bit rate.

Transmission impairment

- Attenuation, distortion, and noise can impair a signal.
- Attenuation is the loss of a signal's energy due to the resistance of the medium.
- Distortion is the alteration of a signal caused by differing propagation speeds of the frequency components.
- Noise is external energy that corrupts a signal.

Appendix: Terms I

- **Analog signal:** continuous signal that can take any value in a range
- **Digital signal:** discrete signal that can take only specific values
- **Periodic signal:** signal that repeats itself after a fixed interval (period)
- **Frequency:** number of complete cycles per unit time
- **Bandwidth:** range of frequencies in a signal
- **Attenuation:** loss of signal strength over distance
- **Noise:** unwanted signal that corrupts the original signal
- **SNR:** signal-to-noise ratio; higher is better

Appendix: Terms II

Term	Meaning used in this topic
Composite signal	Signal made up of multiple frequency components
Fourier analysis	Mathematical technique to decompose signals into frequency components
Nyquist formula	Maximum bit rate for a noiseless channel
Shannon formula	Maximum bit rate for a noisy channel
Bit rate	Number of bits transmitted per second
Baud rate	Number of signal level changes per second

Appendix: Terms III

Term	Meaning used in this topic
Phase	Position of waveform relative to time zero
Propagation time	Time taken for a signal to travel from source to destination
Transmission time	Time taken to transmit all bits of a message
Bandwidth-delay product	Product of bandwidth and propagation time; represents bits in the link
Distortion	Alteration of signal due to different propagation speeds of frequency components

Appendix: Sources

Slide portion	Primary source	Source pages used
Analog vs. digital signals, periodic signals, frequency, phase	Forouzan	Forouzan Ch. 3 pp. 46–67
Composite signals, bandwidth, Fourier analysis	Forouzan	Forouzan Ch. 3 pp. 67–82
Digital signals, bit rate, baud rate	Forouzan	Forouzan Ch. 3 pp. 82–95
Transmission impairment: attenuation, distortion, noise	Forouzan	Forouzan Ch. 3 pp. 95–110
Data rate limits: Nyquist & Shannon formulas	Forouzan	Forouzan Ch. 3 pp. 110–125
Performance: throughput, latency, bandwidth-delay product	Forouzan	Forouzan Ch. 3 pp. 125–135

Questions?